Lesson Notes

MODULE 3 | LESSON 2

CRITICAL LINE ALGORITHM AND CORNER PORTFOLIOS

Reading Time	90 minutes
Prior Knowledge	Optimization procedures required in obtaining optimal portfolios, Calculus with matrix
Keywords	Critical line algorithm, turning points, corner portfolios

In the previous lesson, we extended the classical mean-variance optimization paradigm to incorporate portfolio managers' views into a model via the Black-Litterman model.

Regardless of what model is used, when inequality constraints are part of the model, we need a reliable tool to compute optimal portfolios. This is the critical line algorithm that will be presented next.

Both in the classical mean-variance optimization model and the Black-Litterman model, one has to resort to some kind of portfolio optimization. In its essence, this amounts to identifying the most suitable portfolio on the efficient frontier given a certain set of constraints.

The constraints change depending on the particular portfolio manager, their tactical portfolio choices, target returns, or threshold volatility assigned to the manager by the clients. In some cases, one may want to hold a position in a specific asset because such an asset is recognized as having strong upside, but its lack of liquidity may suggest to limit the exposure of the portfolio to this particular asset.

When we start to have inequality constraints in our portfolio, we are already aware that there is no closed form solution, unlike the case when we have only equality constraints. While there are packages that can compute numerical solutions for most feasible problems, Markowitz himself developed a method to compute a solution, and his method is called the **critical line algorithm** (CLA). Indeed, CLA has become a favorite among practitioners because it is arguably the only algorithm specifically designed for inequality-constrained portfolio optimization, which guarantees that an exact solution will be identified in a finite number of iterations.

Why can we not rely on the other numerical methods? Well, we can, but gradient-based algorithms depend on a seed vector of initial values and could converge to local maxima or minima. Not so for the CLA. In addition, gradient-based algorithms identify only the optimal solution (assuming it is a global solution and not a local solution, which would obviously be sub-optimal); on the other hand, the CLA does not find a single portfolio and actually derives the entire efficient frontier.

It is worthwhile to keep in mind that CLA was developed by Markowitz to handle optimizing quadratic functions subject to linear inequality constraints. If, for any reason, one found oneself working with nonlinear inequality constraints, the methodology would not work.

This paper provides a step-by-step implementation of the CLA in the Python language. As the paper intersperses theory and coding, we will cover some important points separately in the notes to make sure that *you will not miss the forest for the trees*.

Before starting the lesson, students should be warned that a mere copy pasting of code in the paper to a Jupyter notebook or any other IDE may not work smoothly as the code in the paper may contain additional spaces which are likely to produce error messages when the code is required to run in Python.

1. Set up and essence of the CLA

We assume that we have n assets, and therefore, we have an $n \times 1$ vector of means (expected returns or, in the parlance of the Black-Litterman model, equilibrium excess returns or posterior excess returns) and an $n \times n$ variance covariance matrix (which could be the posterior variance covariance matrix Σ_{BL} if working in the Black-Litterman model setting). As before, we need to assume that these parameters are time homogeneous, i.e., they are constant over time (at least for the period of time that is of relevance to us). So far, nothing is different from what we did before.

Now, we need to introduce some additional elements. Specifically:

- $N = \{1, 2, \dots, n\}$ is a set of indices associated with the individual assets in our universe;
- $w_i, i \in N$ denotes the weights of each asset in the portfolio.$w = (w_1, w_2, \dots, w_n)^T$ is the $n \times 1$ vector of weights;
- $l = (l_1, l_2, \dots, l_n)^T$ is an $n \times 1$ vector of lower bounds for the weights.$u = (u_1, u_2, \dots, u_n)^T$ is an $n \times 1$ vector of upper bounds for the weights. Hence, we have that

$$l_i \leq w_i \leq u_i, \quad i \in N;$$

- $F \subset N$ is a subset that collects the free assets, i.e., assets for which $l_i < w_i < u_i$ for $i \in F$;
- $B \subset N$ is a subset that collects the assets whose weights lie on one of the bounds, i.e., either $w_i = l_i$ or $w_i = u_i$;
  \item Clearly $B \cup F = N$, i.e. any asset belongs to either F or B;
- Weights are required to sum up to 1, and the portfolio must achieve the required mean return,r_p, (assuming we solve the problem by minimizing the variance of the portfolio that achieves the minimum required return).

We need to maintain a distinction between free assets' indices and assets whose weights lie on one of their bounds. For this purpose, we will decompose the mean return vector and the variance covariance matrix of the assets as follows:

$$\mu = \begin{pmatrix} \mu_F \\ \mu_B \end{pmatrix}, \quad \text{and} \quad \Sigma = \begin{pmatrix} \Sigma_F & \Sigma_{FB} \\ \Sigma_{BF} & \Sigma_B \end{pmatrix}$$

where Σ_F is a $k \times k$ matrix,k being the number of free assets,Σ_B an $(n-k) \times (n-k)$ matrix, Σ_{FB} is $k \times (n-k)$, and Σ_{BF} is $(n-k) \times k$. The vector of portfolio weights is partitioned accordingly, i.e.,$w^T = (w_F^T, W_B^T)$. Finally, note also that $\Sigma_{FB} = \Sigma_{BF}^T$.

The key technical element for CLA is the concept of **turning point**. The definition of turning point is a practical one: a vector w^* is a turning point if, close to it, there is another solution vector with different free assets, i.e., at least one of the weights previously free becomes bounded and/or one of the previously bounded weights becomes free. This is important. Say that $W \subset R^N$ is the region of the Euclidean space R^N where our weights are allowed to exist. Then, in the regions of W away from turning points, the inequality constraints are for practical purposes irrelevant with respect to the free assets. Since the free weights are, as the name implies, free to move in that part of the region, this is equivalent to solving an unconstrained problem like (3) in the required reading, i.e.:

$$L(w_F, w_B, \gamma, \lambda) = \frac{1}{2} w_F^T \Sigma_F w_F + \frac{1}{2} w_F^T \Sigma_{FB} w_B + \frac{1}{2} w_B^T \Sigma_{BF} w_F + \frac{1}{2} w_B^T \Sigma_B w_B \\ - \gamma(w_F^T \iota_k + w_B^T \iota_{n-k} - 1) - \lambda(w_f^T \mu_F + w_B^T \mu_B - \mu_p)$$

with ι_k a $k \times 1$ vector of ones and ι_{n-k} an $(n-k) \times 1$ vector of ones. The nature of the Lagrangian function should be familiar by now. The thing to keep in mind is that, now,w_B is constrained and thus does not change between turning points. Hence, we need to find only the optimal $w_F^\star$ and since, again in between turning points, the w_F's are not constrained (in the sense that the w_F's are not subject to inequality constraints, more properly), this is an unconstrained optimization problem so to speak.

Markowitz's CLA works in the following way:

- Identify the turning points.
- At each turning point (pay attention since this means that some of the weights will be set equal to some value based on their active constraint), compute the optimal portfolio.
- The efficient frontier is derived as the convex combination between any two neighboring turning points (similar to what we did when using the two fund theorem in a previous lesson).

How do we find the turning points?

The algorithm requires an initial solution on the constrained minimum variance frontier, and it starts by identifying the turning point with the highest expected return. Then, we move to the next lower turning point.

Step 1. Order all assets with respect to their expected return values. Then, the first assets have the highest expected return and the n-th assets the lowest expected return.

Step 2. Set all weights initially at their lower bounds,$w_i = l_i$.

Step 3. Increase the weights of the first asset. If the upper bound is reached and $w^T \iota < 1$, increase the second asset's weight and so forth. This part terminates when $w^T \iota = 1$. Note that this requires that $w^T l \leq 1 \leq w^T u$ or there would be no solution! In particular, we stop for the asset whose index number k is such that

$$\sum_{i=1}^{k} u_i + \sum_{j=k+1}^{n} l_j \leq 1 \quad \text{and}$$

$$\sum_{i=1}^{k+1} u_i + \sum_{j=k+2}^{n} l_j > 1.$$

Step 4. The solution is a vector of weights where the weights of the first assets are set to the upper bounds, the last assets' weights are set to their lower bounds, and there is an asset in the middle part whose weight is between its lower and upper bounds. Specifically, the weight of the k-th asset (now they are ordered) is going to be such that $l_k < 1 - \sum_{i=1}^{k-1} u_i - \sum_{j=k+1}^{n} l_i$. This particular asset (k) is the only one that is not set to either its upper or lower bound, and it is called the **free asset**. For convenience, we can relabel the index k as i_f. Then

$$w_{i_f} = 1 - \sum_{j \in U} w_j - \sum_{k \in L} w_k = 1 - w_B^T \iota_{n-k}.$$

Pay attention that at this first stage $F = \{i_f\}$ and $B = N \setminus \{i_f\}$.

Step 5. In going from one turning point to the next, it is necessary that one element is either added to the list of free assets,F, or removed from it. This can be done by changing the value of λ in $L(w_F, w_B, \gamma, \lambda)$ as defined above, letting it become smaller and smaller. This is not immediately obvious, and it happens because λ and $w^T\mu$ are linearly and positively related. (This is not really dealt with in the required reading). If we differentiate $L(w_F, w_B, \gamma, \lambda)$ with respect to w_F and set that system of equations to zero, we get

$$\Sigma_F w_F + \Sigma_{FB} w_B - \lambda \mu_F = \gamma \iota_F$$

where ι_F is a vector of 1s of length equal to the number of elements in F. Now, using the condition that the weights must sum to one, we have also

$$w_F^T \iota_F = 1 - w_B^T \iota_B.$$

Solving the last two systems of equations together with respect to γ implies that

$$\gamma = -\lambda \frac{\iota_F^T \Sigma_F^{-1} \mu_F}{\iota_F^T \Sigma_F^{-1} \iota_F} + \frac{1 - \iota_B^T w_B + \iota_F^T \Sigma_F^{-1} \Sigma_{FB} w_B}{\iota_F^T \Sigma_F^{-1} \iota_F}.$$

Also, we have easily determined that

$$w_F = -\Sigma_F^{-1} \Sigma_{FB} w_B + \gamma \Sigma_F^{-1} \iota_F + \lambda \Sigma_F^{-1} \mu_F.$$

Then, we have

$$\mu^T w = \mu_F^T w_F + \mu_B^T w_B = -\mu_F^T \Sigma_F^{-1} \Sigma_{FB} w_B + \gamma H + \lambda K$$

where

$$H = \iota_F^T \Sigma_F^{-1} \mu_F, \ \ K = \mu_F^T \Sigma_F^{-1} \mu_F.$$

Differentiating the last expression with respect to λ and using the expression for γ found above, we get

$$\frac{\partial \mu^T w}{\partial \lambda} = K - \frac{H^2}{M}$$

with $M = \iota_F^T \Sigma_F^{-1} \iota_F$. At this point, we know that between turning points Σ_F does not change and therefore $\mu^T w$ is a linear function of λ and the slope is given by $K - \frac{H^2}{M}$. Furthermore, one can show that this slope is positive (it is a bit tedious, but it is only algebra).

This linear relationship between λ and $\mu^T w$ implies that every subsequent turning point will have a lower value for λ. In this particular step, our F has only one asset and we have only the option to add another asset to F. However, in general, in the steps that will follow, moving from turning point to turning point by decreasing λ, we can have one of the following two situations:

1. either one free assets moves to one of its bounds or
2. an asset that was previously on one of its bounds becomes free.

These two cases must be considered to compute the λ and w associated with each of the turning points. To this purpose, the required paper deals with two separate cases.

(Case 1: Free asset moves to its bound). In this case, suppose that λ_t is associated with the current turning point and let F be the set of free assets slightly below this turning point, which is the same as selecting a$\lambda_{t+1} < \lambda < \lambda_t$. The index t refers to the successive values of λ associated with a turning point (hence the index "t"). Exploiting the documented linear relationship between λ and $\mu^T w$, if we decrease λ, asset i will hit the lower bound if $dw_i/d\lambda > 0$, and the upper bound whenever $dw_i/d\lambda < 0$.

If $\lambda^{(i)}$ is the λ associated with the turning point where asset i hits one of the bounds, then one could show that its formula is given by

$$\lambda^{(i)} = \frac{1}{C_i}\left[\left(1 - \iota_B^T w_B + \iota_F^T \Sigma_F^{-1} \Sigma_{FB} w_B\right)(\sigma_F^{-1}\iota_F)_i - (\iota_F^T(\Sigma_F^{-1}\iota_F)_i(b_i + (\Sigma_F^{-1}\Sigma_{FB}w_B)_i\right]$$

where

$$C_i = -(\iota_F^T \Sigma_F^{-1} \iota_F)(\Sigma_F^1 \mu_F)_i + (\iota_F^T \Sigma_F^{-1} \mu_F)(\Sigma_F^{-1}\iota_F)_i$$

and

$$b_i = \begin{cases} u_i & \text{if } C_i > 0; \\ l_i & \text{if } C_i < 0. \end{cases}$$

This case should be used only when we have at least two free assets since for $k = 1$ $C_i = 0$.

Then, the next $\lambda < \lambda_t$ where an asset leaves the subset F is given by

$$\lambda_{out} = \max_{i \in F}\{\lambda^{(i)}\}.$$

(Case 2. One asset is on its bound and becomes free). When we decrease λ, it is possible that an asset i that was on its bound becomes free. The corresponding portfolio is therefore a turning point, and the subsets F and B must be redefined:\

$$F_i = F \cup \{i\}, \quad B_i = B \setminus \{i\}, \; i \in B.$$

In this case, the formula for λ_i is like the one before, but we have to replace B with B_i and F with F_i just defined. That is:

$$\lambda^{(i)} = \frac{1}{C_i}\left[\left(1-\iota_{B_i}^T w_{B_i} + \iota_{F_i}^T \Sigma_{F_i}^{-1}\Sigma_{F_i B_i} w_{B_i}\right)(\sigma_{F_i}^{-1}\iota_{F_i})_i - (\iota_{F_i}^T \Sigma_{F_i}^{-1}\iota_{F_i})_i (b_i + (\Sigma_{F_i}^{-1}\Sigma_{F_i B_i} w_{B_i})_i\right]$$

where

$$C_i = -(\iota_{F_i}^T \Sigma_{F_i}^{-1}\iota_{F_i})(\Sigma_{F_i}^{1}\mu_{F_i})_i + (\iota_{F_i}^T \Sigma_{F_i}^{-1}\mu_{F_i})(\Sigma_{F_i}^{-1}\iota_{F_i})_i$$

and

$$b_i = \begin{cases} u_i & \text{if asset i was previously on its upper bound} \\ l_i & \text{if asset i was previously on its lower bound.} \end{cases}$$

NOTE: These results for λ_i, C_i, b_i are derived by starting from the equation

$$w_F = -\Sigma_F^{-1}\Sigma_{FB}w_B + \gamma\Sigma_F^{-1}\iota_F + \lambda\Sigma_F^{-1}\mu_F,$$

replacing γ with its values derived earlier, which yields an expression that contains only λ and many constants. Then, one takes the derivatives of the i-th components with respect to λ and perform algebra on the equations setting the bound equal to l_i or b_i as needed. It is not particularly difficult, but it is clearly tedious work.

Define

$$\lambda_{in} = \max_{i\in B}\{\lambda^i : \lambda^i < \lambda_t\}$$

and if there is no $\lambda^i < \lambda_t$, there is no solution for λ_{in}.

Step 6. To identify the next turning point, we use the values found for λ_{in} and λ_{out}. We define

$$\lambda_{new} \equiv \lambda_{t+1} = \max\{\lambda_{in}, \lambda_{out}\}$$

If there is only a solution for λ_{in} or λ_{out}, then λ_{new} is overwritten by the respective value.

Recall that λ_{in} means that for that λ asset i leaves B and enters in F while λ_{out} is the value of λ for which asset i enters B. Hence, we need to keep this in mind when using λ_{new}.

Depending on which case occurs we have to replace F by $F \setminus \{i\}$ or $F_i = F \cup \{i\}$, or $B = B \cup \{i\}$ or $B_i = B \setminus \{i\}$. We must be sure to remember to update the current value for λ.

Step 7. Once we reach a situation where there is no longer a solution for λ_{in} and λ_{out}, the algorithm is completed as we have reached the lower turning point.

At this point, a student should be in a position to read the paper and make sense of the coding done there. Once properly implemented, it will be a working algorithm.

2. Corner portfolios

We will discuss an example where short selling is constrained (i.e.$l_i = 0$ for $i = 1, 2, \ldots, n$). In addition, weights are expected to sum to 1.00, as always.

We are familiar with the two fund (or two portfolio) theorem; in the case we have constraints, it is not difficult to prove that combining two neighboring turning points linearly also leads to a constrained minimum variance portfolio. This relies on the fact that by writing the function $L(w_F, w_B, \gamma, \lambda)$ the way we did, we are actually solving unconstrained optimization problems (where some of the weights are kept fixed at one of their bounds) and thus a linear combination of two solutions of the unconstrained problem is itself a solution.

The optimal portfolios associated with each turning point are called **corner portfolios**. Then, the efficient frontier can be closely approximated using linear combinations of a small number of corner portfolios.

Given five assets, a portfolio manager who only has no short selling constraints has run the critical line algorithm and has obtained the following output.

The typical output using CLA is presented in Figure 1. Note that in this example 6 corner portfolios are identified and that the portfolio compositions are to be read along the rows.

Figure 1: Example of Output Using CLA .

				Asset Weights				
Corner P.folio	Expected Return	Expected Std. dev.	Sharpe Ratio	Asset 1	Asset 2	Asset 3	Asset 4	Asset 5
1	15%	18.5%	0.81	0.00	0.00	1.00	0.00	0.00
2	13.5%	16.2%	0.83	0.00	0.00	0.85	0.00	0.15
3	13.1%	13.6%	0.96	0.22	0.00	0.55	0.00	0.23
4	12.6%	13.0%	0.97	0.21	0.00	0.50	0.05	0.24
5	10.6%	11.0%	0.91	0.10	0.49	0.21	0.00	0.20
6	9.5%	10.9%	0.87	0.00	0.87	0.04	0.00	0.09

The client, a pension fund, has stated to the portfolio manager that they would need a return to match the current spending rate of $8.5\%,$ inflation (estimated at 2.2%) and cost of managing the pension fund (expected to be $0.50\%.$) The trustees of the pension fund have stated that they would like to preserve the purchasing power of the fund, and they also explained to the portfolio manager that the standard deviation should be no more than 12.0% a year.

Then, the required rate of return that the portfolio manager needs to achieve is:

$$(1 + 8.5\%)(1 + 2.2\%)(1 + 0.50\%) - 1 = 11.44\%.$$

What should the portfolio manager do based on the table they received from their quantitative analyst?

We notice that the required rate of return lies between Corner Portfolio 4 and Corner Portfolio 5. Hence, we consider a linear combination of these two corner portfolios, i.e.:

$$12.6\%\alpha + 10.6\%(1 - \alpha) = 11.44\%.$$

This means that $2\alpha = 0.84$ or $\alpha = 0.42.$ Therefore, the manager will invest 0.42 in CP 4 and 0.58 in CP 5. What does this mean in practice in terms of the weights for each of the five asset classes?

It's simple:

$$\text{Asset 1} : 0.42 * (0.21) + 0.58 * (0.10) = 0.146;$$

$$\text{Asset 2} : 0.42 * (0.00) + 0.58 * (0.49) = 0.284;$$

$$\text{Asset 3} : 0.42 \times (0.50) + 0.58 \times (0.21) = 0.332;$$

$$\text{Asset 4} : 0.42 \times (0.05) + 0.58 \times (0.00) = 0.021;$$

$$\text{Asset 5} : 0.42 \times (0.24) + 0.58 \times (0.20) = 0.217.$$

Note that the sum of the assets weights is equal to 1 as we would expect.

The portfolio manager was also assigned a constraint on the max volatility the pension fund is willing to accept. The volatility is computed using the volatilities of the corner portfolios. Since we do not know the correlation between the two corner portfolios, the standard deviation of this portfolio can be closely approximated by the weighted average of the two relevant corner portfolios. In this case, we have

$$\sigma_p = 0.42 \times 13\% + 0.58 \times 11\% = 11.84\%$$

which is less than 12% and hence the portfolio satisfies the risk-requirement as well.

The Sharpe ratio for this portfolio (assuming a risk-free interest rate of zero) is 0.966. The Board of Trustees is quite happy with the proposal they receive from the portfolio manager. Out of curiosity, they inquire what the Sharpe ratio is for the market. This would be the Sharpe ratio of the corner portfolio with the highest Sharpe ratio, i.e., 0.97. The trustee are now really happy with their choice of the portfolio manager.

The last point about identifying the market portfolio, i.e., the corner portfolio with the highest Sharpe ratio is quite important. In fact, if there were a risk-free asset and we are allowed to borrow, we could use the CAL line as we did before. But, for that, we need to be able to identify the market portfolio, which we just did.

3. Conclusion

In this lesson, we learned how to solve portfolio optimization problems (both the Markowitz and the Black-Litterman) when inequality constraints are required. Furthermore, we investigated the notion of corner portfolios and how they can be used in real-life applications.

In the next lesson, we'll consider the Black-Litterman model again. However, this time, we will look at what happens when practitioners are tempted to take certain shortcuts.